



**CURIOSITY
CREATES**

**A BETTER
WORLD**

**TOPIC 4 –
BONDING THEORY**

QEng Prep - Chemistry



FACULTY OF
ENGINEERING AND
APPLIED SCIENCE

KEY CONCEPTS

The key concepts covered in this lecture include:

- Predict the shapes of simple molecules and ions (e.g., CH_4 , SO_3 , O_2 , H_2O , NH_4^+), using Lewis structure models and the valence shell electron pair repulsion (VSEPR) model;
- Draw diagrams to represent simple molecules and ions (e.g., CH_4 , SO_3 , O_2 , H_2O , NH_4^+); and
- Predict the polarity of various chemical compounds, based on their molecular shapes and the difference in the electronegativity values of the atoms.

LEWIS STRUCTURES

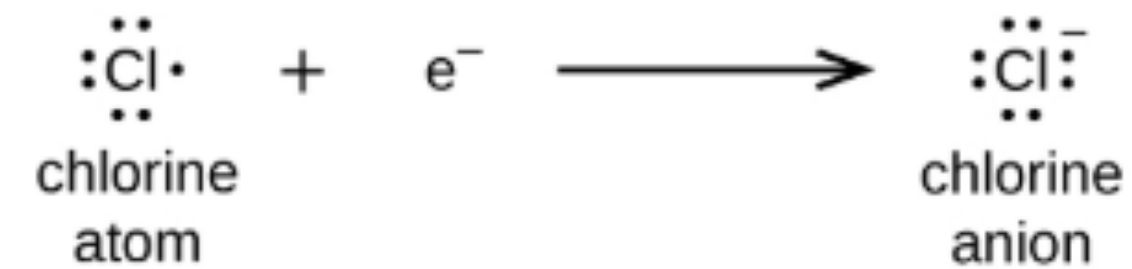


- We use Lewis symbols to describe valence electron configurations of atoms and monatomic ions. A Lewis symbol consists of an elemental symbol surrounded by one dot for each of its valence electrons:



Atoms	Electronic Configuration	Lewis Symbol
sodium	$[\text{Ne}]3s^1$	$\text{Na} \cdot$
magnesium	$[\text{Ne}]3s^2$	$\cdot \text{Mg} \cdot$
aluminum	$[\text{Ne}]3s^2 3p^1$	$\cdot \text{Al} \cdot$
silicon	$[\text{Ne}]3s^2 3p^2$	$\cdot \ddot{\text{Si}} \cdot$
phosphorus	$[\text{Ne}]3s^2 3p^3$	$\cdot \ddot{\text{P}} \cdot$
sulfur	$[\text{Ne}]3s^2 3p^4$	$:\ddot{\text{S}} \cdot$
chlorine	$[\text{Ne}]3s^2 3p^5$	$:\ddot{\text{Cl}} \cdot$
argon	$[\text{Ne}]3s^2 3p^6$	$:\ddot{\text{Ar}}:$

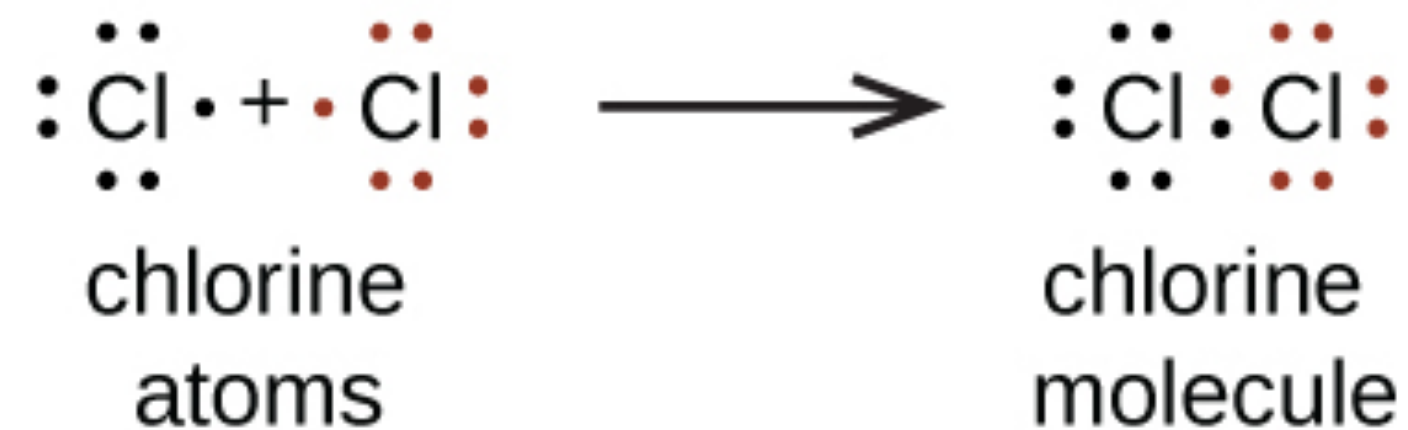
- Lewis symbols can also be used to illustrate the formation of cations and anions from atoms, as shown here for sodium and chlorine:



LEWIS STRUCTURES

General Overview

- We also use Lewis symbols to indicate the formation of covalent bonds, which are shown in Lewis structures, drawings that describe the bonding in molecules and polyatomic ions. For example, when two chlorine atoms form a chlorine molecule, they share one pair of electrons:



- The Lewis structure indicates that each Cl atom has three pairs of electrons that are not used in bonding (called lone pairs) and one shared pair of electrons (written between the atoms). A dash (or line) is sometimes used to indicate a shared pair of electrons.

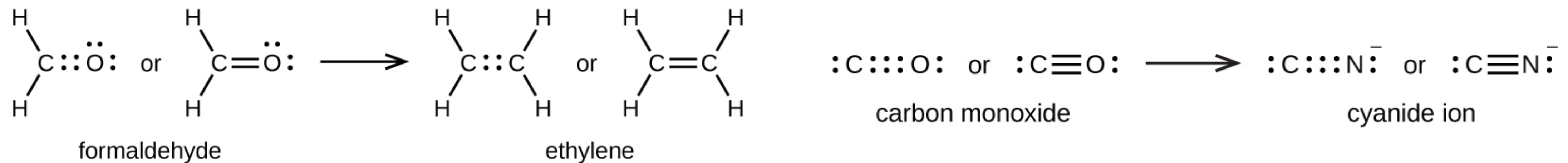
LEWIS STRUCTURES

The Octet Rule

- Other halogen molecules (F_2 , Br_2 , I_2 , and At_2) form bonds like those in the chlorine molecule: one single bond between atoms and three lone pairs of electrons per atom. This allows each halogen atom to have a noble gas electron configuration. The **tendency of main group atoms to form enough bonds to obtain eight valence electrons** is known as the octet rule.
- The number of bonds that an atom can form can often be predicted from the number of electrons needed to reach an octet (eight valence electrons); this is especially true of the nonmetals of the second period of the periodic table (C, N, O, and F).

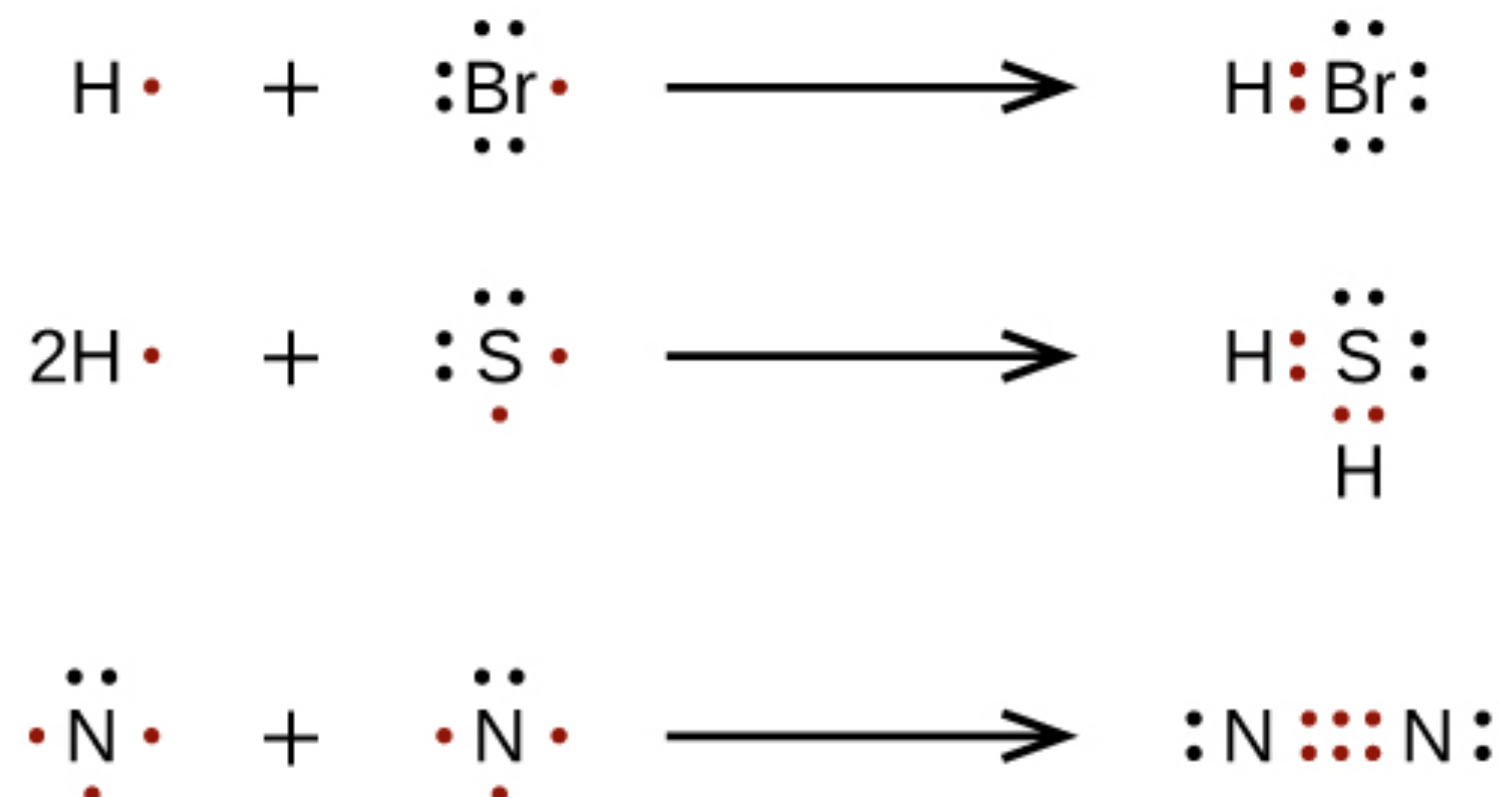
LEWIS STRUCTURES

- Double and triple bonds can be illustrated using Lewis structures as well:



Writing Lewis Structures with the Octet Rule

- For very simple molecules and molecular ions, we can write the Lewis structures by merely pairing up the unpaired electrons on the constituent atoms:



- For more complicated molecules and molecular ions, it is helpful to follow the step-by-step procedure outlined on the next slide.

LEWIS STRUCTURES

Writing Lewis Structures with the Octet Rule

1. Determine the total number of valence (outer shell) electrons. For cations, subtract one electron for each positive charge. For anions, add one electron for each negative charge.
2. Draw a skeleton structure of the molecule or ion, arranging the atoms around a central atom. (Generally, the least electronegative element should be placed in the center). Connect each atom to the central atom with a single bond (one electron pair).
3. Distribute the remaining electrons as lone pairs on the terminal atoms (except hydrogen), completing an octet around each atom.
4. Place all remaining electrons on the central atom.
5. Rearrange the electrons of the outer atoms to make multiple bonds with the central atom in order to obtain octets wherever possible.

Examples:

1. Draw a Lewis structure diagram for SiH_4
2. Draw a Lewis structure diagram for CHO_2^- .
3. Draw a Lewis structure diagram for OF_2 .
4. Draw a Lewis structure diagram for NO^+ .

LEWIS STRUCTURES

Examples:

1. Draw a Lewis structure diagram for SiH_4

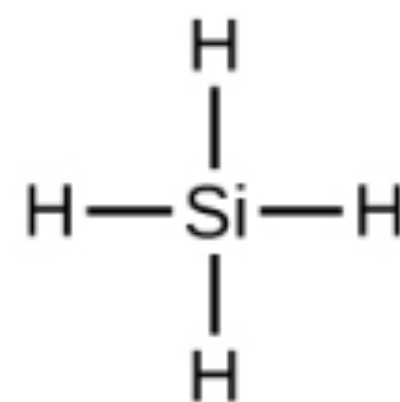
For a molecule, we add the number of valence electrons on each atom in the molecule:

Si: 4 valence electrons/atom x 1 atom = 4

H: 1 valence electron/atom x 4 atoms = 4

Total: 8 valence electrons

Draw a skeleton structure of the molecule:



Distribute the remaining electrons as lone pairs on the terminal atoms (except hydrogen) to complete their valence shells with an octet of electrons.

→ There are no remaining electrons on SiH_4 , so it is unchanged.

Place all remaining electrons on the central atom.

→ There are no remaining electrons; we already placed all of the electrons determined in Step 1.

Rearrange the electrons of the outer atoms to make multiple bonds with the central atom in order to obtain octets wherever possible.

→ Si already has an octet, so nothing needs to be done.

LEWIS STRUCTURES

Examples:

2. Draw a Lewis structure diagram for CHO_2^- .

For a negative ion, such as CHO_2^- , we add the number of valence electrons on the atoms to the number of negative charges on the ion (one electron is gained for each single negative charge):

C: 4 valence electrons/atom x 1 atom = 4

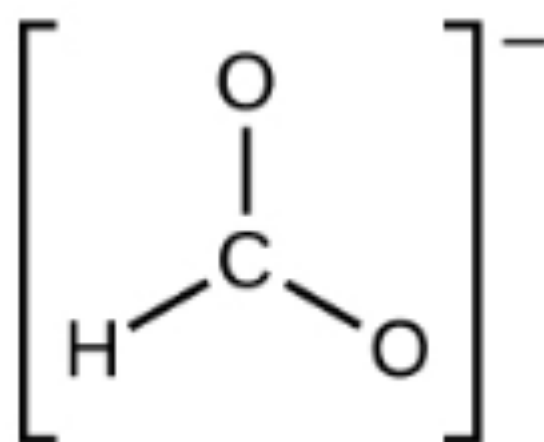
H: 1 valence electron/atom x 1 atoms = 1

O: 6 valence electrons/atm x 2 atoms = 12

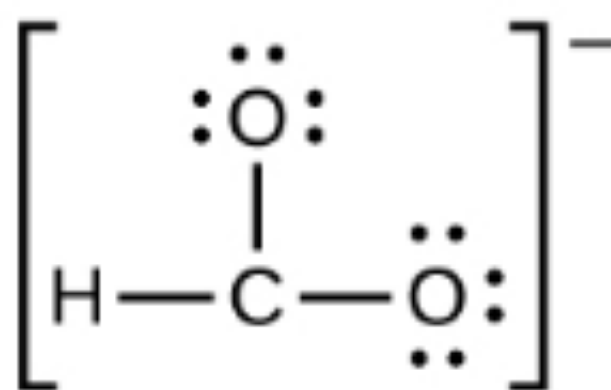
+ 1 additional electron = 1

Total: 18 valence electrons

Draw a skeleton structure of the molecule:



Distribute the remaining electrons as lone pairs on the terminal atoms (except hydrogen) to complete their valence shells with an octet of electrons:



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LEWIS STRUCTURES

Examples:

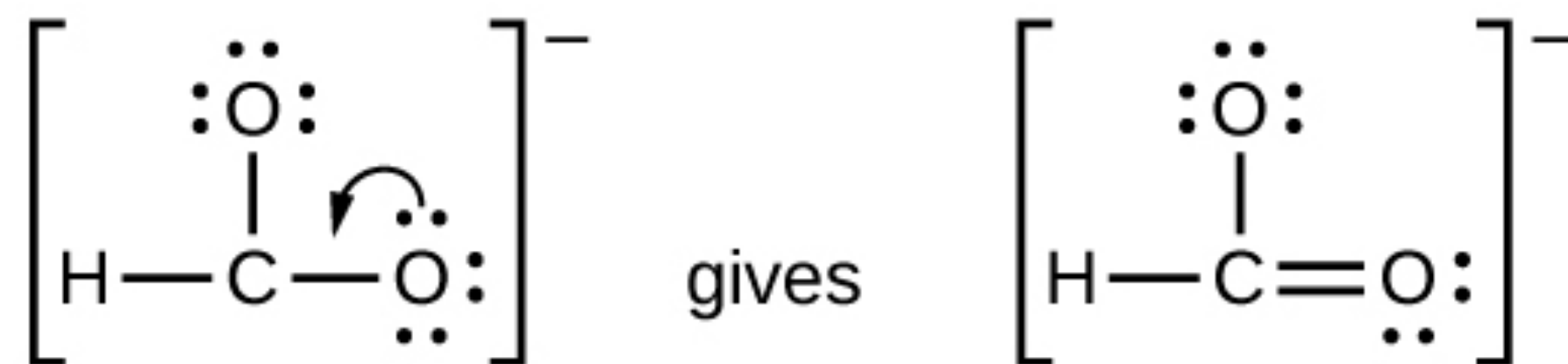
2. Draw a Lewis structure diagram for CHO_2^- (Continued)

Place all remaining electrons on the central atom.

→ There are no remaining electrons; we already placed all of the electrons determined in Step 1.

Rearrange the electrons of the outer atoms to make multiple bonds with the central atom in order to obtain octets wherever possible.

→ We have distributed the valence electrons as lone pairs on the oxygen atoms, but the carbon atom lacks an octet:



LEWIS STRUCTURES

Examples:

3. Draw a Lewis structure diagram for OF_2 .

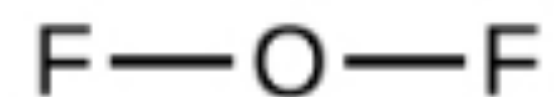
Since OF_2 is a neutral molecule, we simply add the number of valence electrons:

O: 6 valence electrons/atom x 1 atom = 6

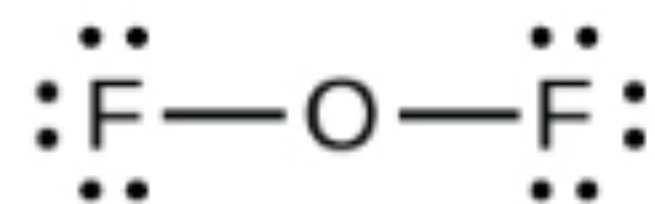
F: 7 valence electrons/atom x 2 atoms = 14

Total: 20 valence electrons

Draw a skeleton structure of the molecule:



Distribute the remaining electrons as lone pairs on the terminal atoms (except hydrogen) to complete their valence shells with an octet of electrons:



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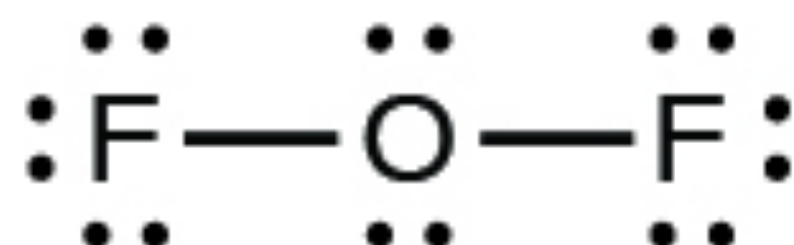
LEWIS STRUCTURES

Examples:

3. Draw a Lewis structure diagram for OF₂ (Continued)

Place all remaining electrons on the central atom.

→ We had 16 electrons remaining in Step 3, and we placed 12, leaving 4 to be placed on the central atom:



Rearrange the electrons of the outer atoms to make multiple bonds with the central atom in order to obtain octets wherever possible.

→ Each atom has an octet as drawn, so nothing changes.

LEWIS STRUCTURES

Examples:

4. Draw a Lewis structure diagram for NO⁺.

For a positive ion such as NO⁺ we add the number of valence electrons on the atoms in the ion and then subtract the number of positive charges on the ion (one electron is lost for each single positive charge) from the total number of valence electrons:

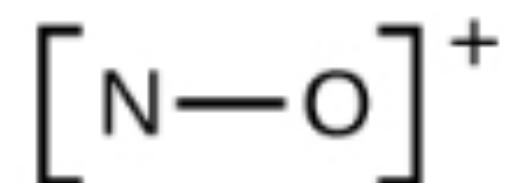
N: 5 valence electrons/atom x 1 atom = 5

O: 6 valence electrons/atom x 1 atom = 6

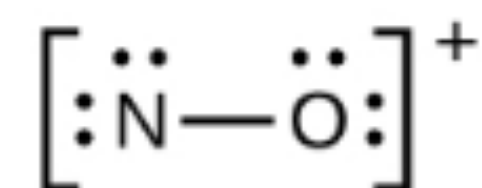
- 1 valence electron (positive charge) = - 1

Total: 10 valence electrons

Draw a skeleton structure of the molecule:



Distribute the remaining electrons as lone pairs on the terminal atoms (except hydrogen) to complete their valence shells with an octet of electrons:



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LEWIS STRUCTURES

Examples:

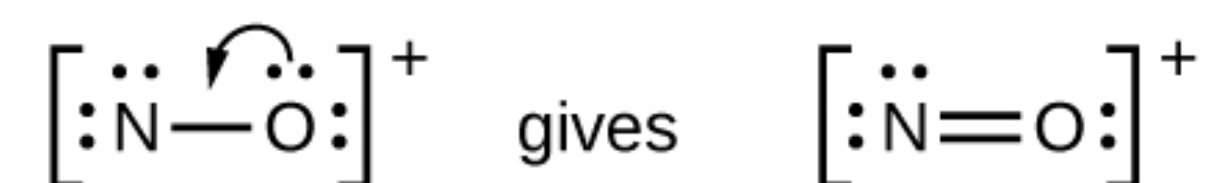
4. Draw a Lewis structure diagram for NO⁺(Continued)

Place all remaining electrons on the central atom.

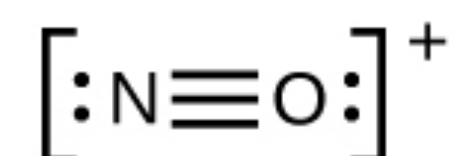
→ There are no remaining electrons; we already placed all of the electrons determined in Step 1.

Rearrange the electrons of the outer atoms to make multiple bonds with the central atom in order to obtain octets wherever possible.

→ For this ion, we added 8 valence electrons, but neither atom has an octet. We cannot add any more electrons since we have already used the total that we found in Step 1, so we must move electrons to form a multiple bond:



→ This still does not produce an octet, so we must move another pair, forming a triple bond:



VALENCE SHELL ELECTRON PAIR REPULSION (VSEPR) MODEL

INTRODUCTION

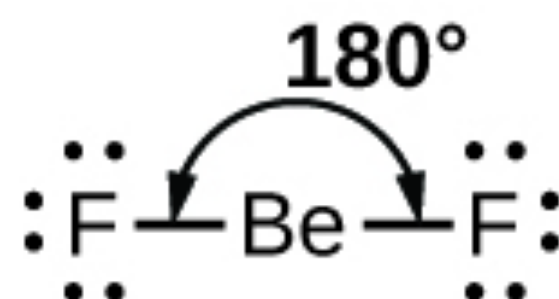
Basic Theory

- VSEPR theory enables us to predict the molecular structure, including approximate bond angles around a central atom, of a molecule from an examination of the number of bonds and lone electron pairs in its Lewis structure.
- The VSEPR model assumes that electron pairs in the valence shell of a central atom will adopt an arrangement that **minimizes repulsions between these electron pairs by maximizing the distance between them.**
- The electrons in the valence shell of a central atom form either bonding pairs of electrons, located primarily between bonded atoms, or lone pairs. The electrostatic repulsion of these electrons is reduced when the various regions of high electron density assume positions as far from each other as possible.

INTRODUCTION

Basic Theory

- As a simple example of VSEPR theory, let us predict the structure of a gaseous BeF_2 molecule. The Lewis structure of BeF_2 shows only two electron pairs around the central beryllium atom:

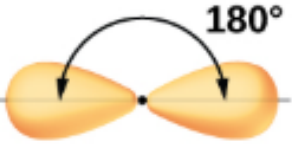
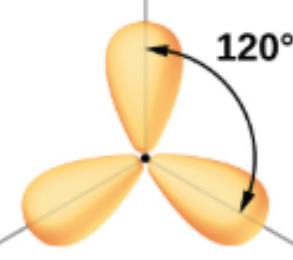
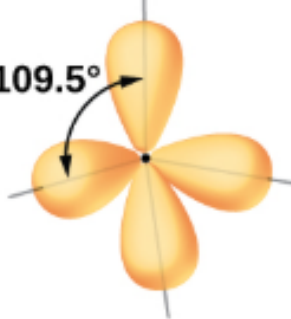
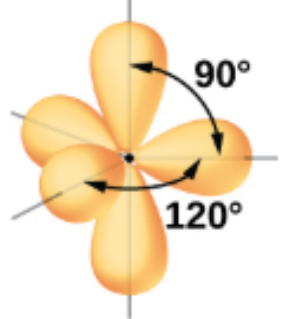
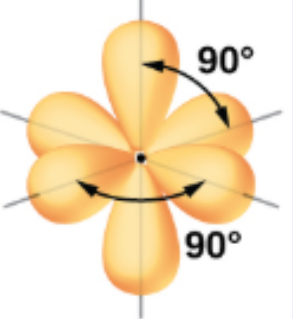


With two bonds and no lone pairs of electrons on the central atom, the bonds are as far apart as possible, and the electrostatic repulsion between these regions of high electron density is reduced to a minimum when they are on opposite sides of the central atom.

The bond angle is 180° . Two regions of electron density around a central atom in a molecule form a **linear** geometry.

INTRODUCTION

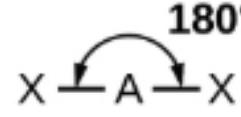
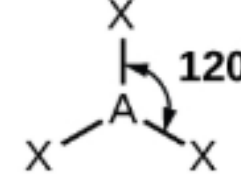
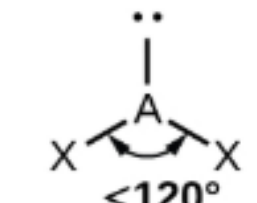
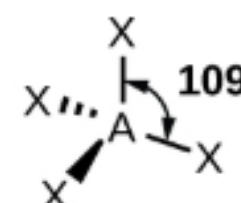
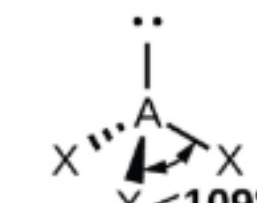
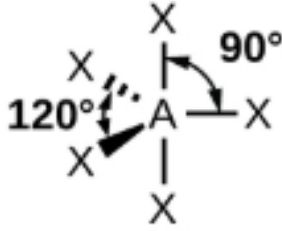
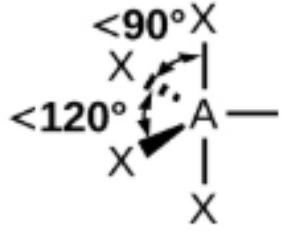
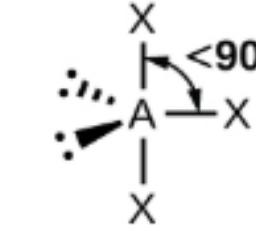
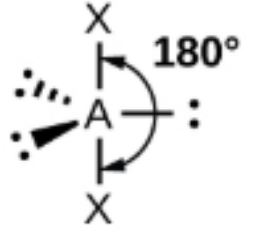
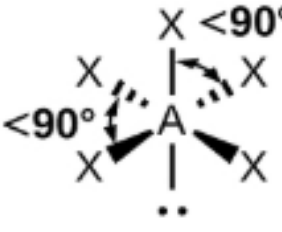
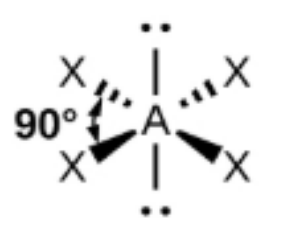
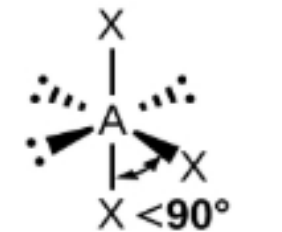
The basic electron-pair geometries predicted by VSEPR theory maximize the space around any region of electron density (bonds or lone pairs) are presented below:

Number of regions	Two regions of high electron density (bonds and/or unshared pairs)	Three regions of high electron density (bonds and/or unshared pairs)	Four regions of high electron density (bonds and/or unshared pairs)	Five regions of high electron density (bonds and/or unshared pairs)	Six regions of high electron density (bonds and/or unshared pairs)
Spatial arrangement					
Line-dash-wedge notation	<chem>H-Be-H</chem>	<chem>[H]B([H])[H]</chem>	<chem>[H]C([H])[H]H</chem>	<chem>F[P](F)(F)F</chem>	<chem>F[S](F)(F)F</chem>
Electron pair geometry	Linear; 180° angle	Trigonal planar; all angles 120°	Tetrahedral; all angles 109.5°	Trigonal bipyramidal; angles of 90° or 120° An attached atom may be equatorial (in the plane of the triangle) or axial (above or below the plane of the triangle).	Octahedral; all angles 90° or 180°

PREDICTING SHAPES USING VSEPR THEORY

The following procedure uses VSEPR theory to determine the electron pair geometries and the molecular structures:

1. Write the Lewis structure of the molecule or polyatomic ion.
2. Count the number of regions of electron density (lone pairs and bonds) around the central atom. A single, double, or triple bond counts as one region of electron density.
3. Identify the electron-pair geometry based on the number of regions of electron density (from table on previous slide).
4. Use the number of lone pairs to determine the molecular structure (from table to the right). If more than one arrangement of lone pairs and chemical bonds is possible, choose the one that will minimize repulsions.

Number of electron pairs	Electron pair geometries: 0 lone pair	1 lone pair	2 lone pairs	3 lone pairs	4 lone pairs
2	 Linear				
3	 Trigonal planar	 Bent or angular			
4	 Tetrahedral	 Trigonal pyramid	 Bent or angular		
5	 Trigonal bipyramid	 Sawhorse or seesaw	 T-shape	 Linear	
6	 Octahedral	 Square pyramid	 Square planar	 T-shape	 Linear

PREDICTING SHAPES USING VSEPR THEORY

Examples:

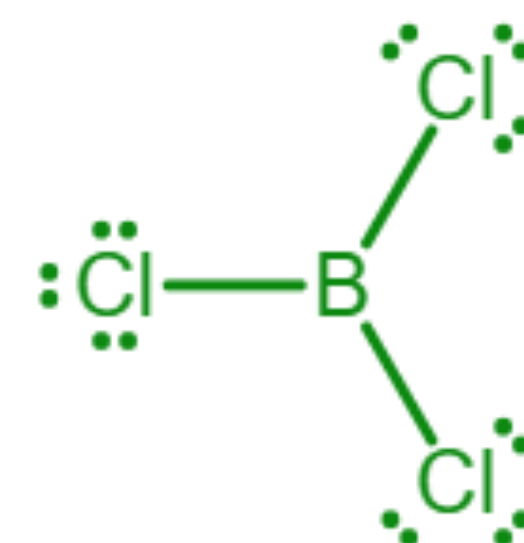
1. Predict the electron-pair geometry and molecular structure of boron trichloride.
2. Predict the electron-pair geometry and molecular structure of the ammonium ion (NH_4^+).
3. Predict the electron-pair geometry and molecular structure of a water molecule.
4. Predict the electron-pair geometry and molecular structure of a SF_4 molecule.

PREDICTING SHAPES USING VSEPR THEORY

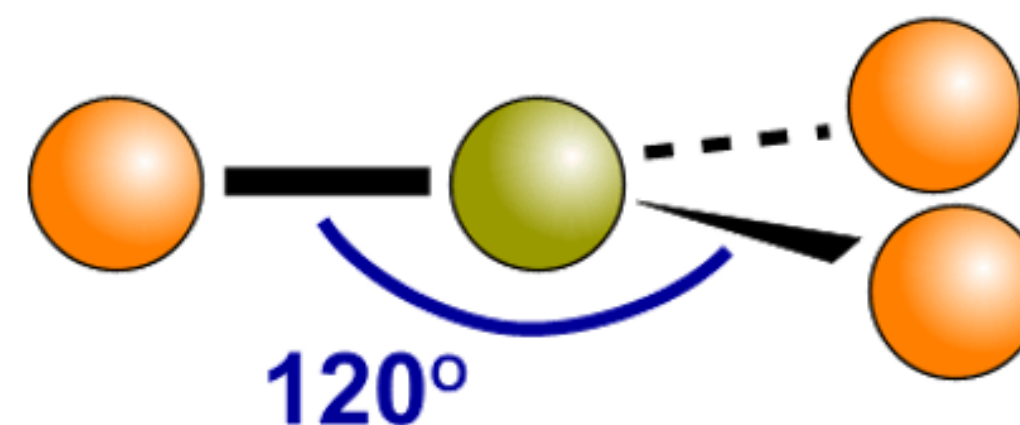
Examples:

1. Predict the electron-pair geometry and molecular structure of boron trichloride.

We write the Lewis structure of BCl_3 as:



We see that BCl_3 contains three bonds, and there are no lone pairs of electrons on boron. The arrangement of three regions of high electron density gives a trigonal planar electron-pair geometry. The B–Cl bonds lie in a plane with 120° angles between them. BCl_3 also has a trigonal planar molecular structure:



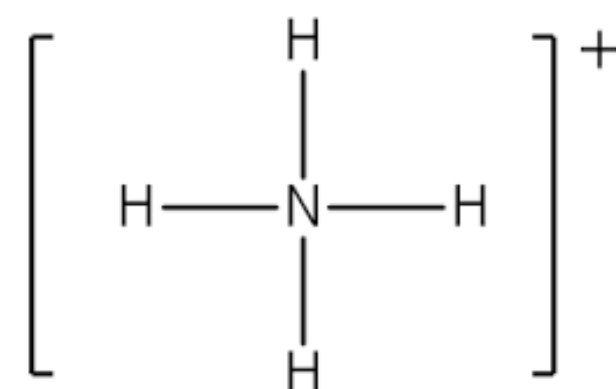
The electron-pair geometry and molecular structure of BCl_3 are both trigonal planar.

PREDICTING SHAPES USING VSEPR THEORY

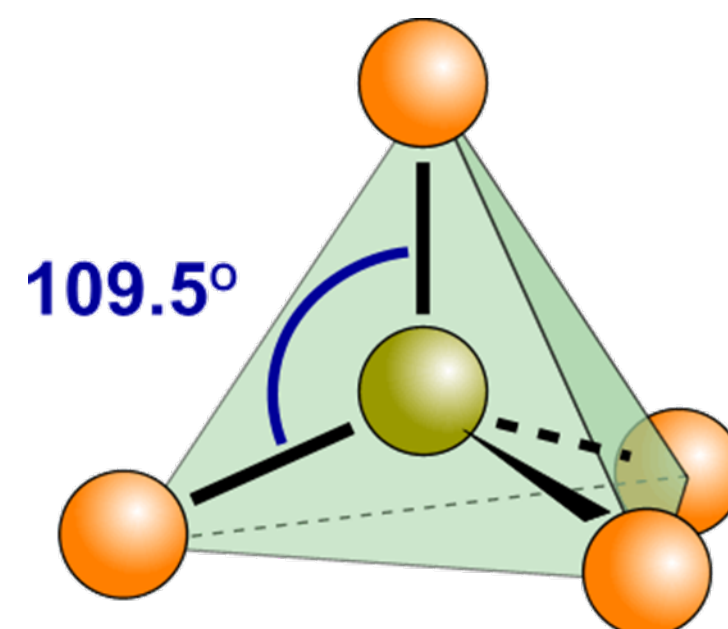
Examples:

2. Predict the electron-pair geometry and molecular structure of the ammonium ion (NH_4^+).

We write the Lewis structure of NH_4^+ as:



We can see that NH_4^+ contains four bonds from the nitrogen atom to hydrogen atoms and no lone pairs. We expect the four regions of high electron density to arrange themselves so that they point to the corners of a tetrahedron with the central nitrogen atom in the middle. Therefore, the electron pair geometry of NH_4^+ is tetrahedral, and the molecular structure is also tetrahedral:

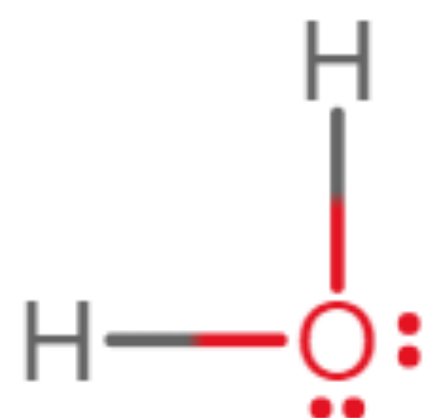


PREDICTING SHAPES USING VSEPR THEORY

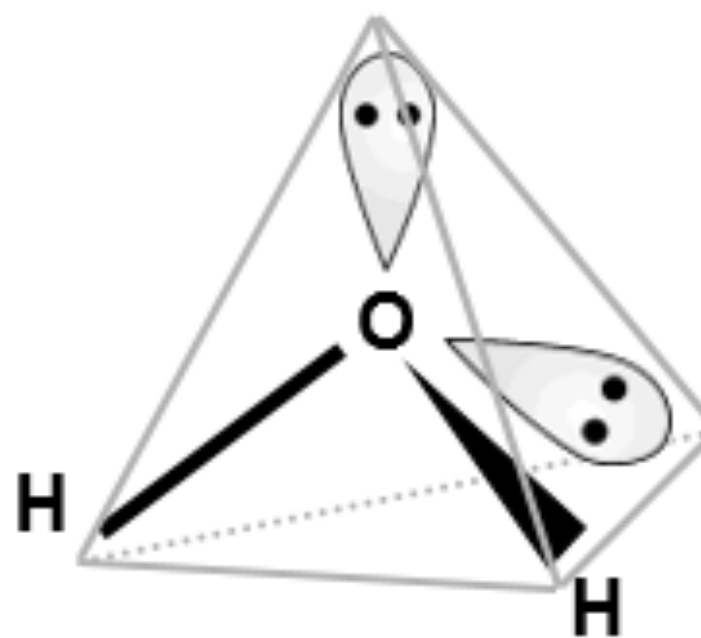
Examples:

3. Predict the electron-pair geometry and molecular structure of a water molecule.

We write the Lewis structure of H_2O as:



We predict that these four regions are arranged in a tetrahedral fashion. As such, the electron-pair geometry is tetrahedral and the molecular structure is bent with an angle slightly less than 109.5° :



PREDICTING SHAPES USING VSEPR THEORY

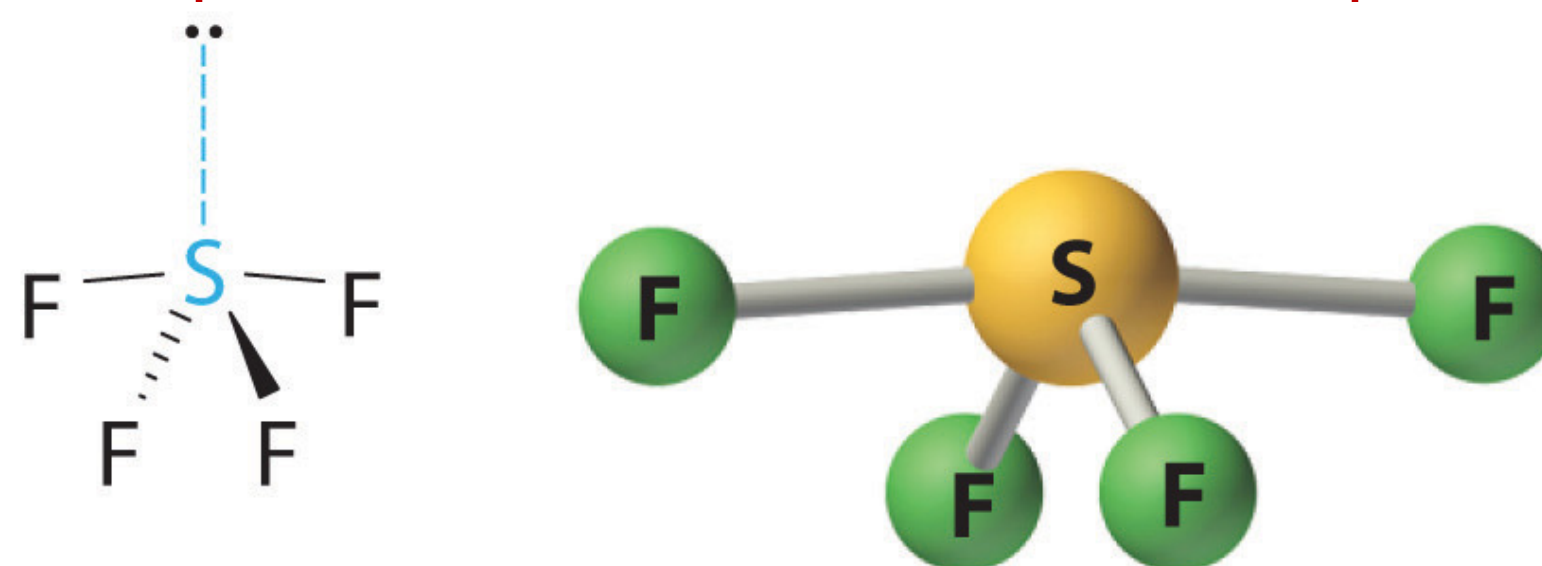
Examples:

4. Predict the electron-pair geometry and molecular structure of a of a SF_4 molecule.

The Lewis structure of SF_4 indicates five regions of electron density around the sulfur atom: one lone pair and four bonding pairs:



We expect these five regions to adopt a trigonal bipyramidal electron-pair geometry. To minimize lone pair repulsions, the lone pair occupies one of the equatorial positions. SF_4 has a trigonal bipyramidal arrangement of the five regions of electron density. One of the regions is a lone pair, which results in a seesaw-shaped molecular structure:

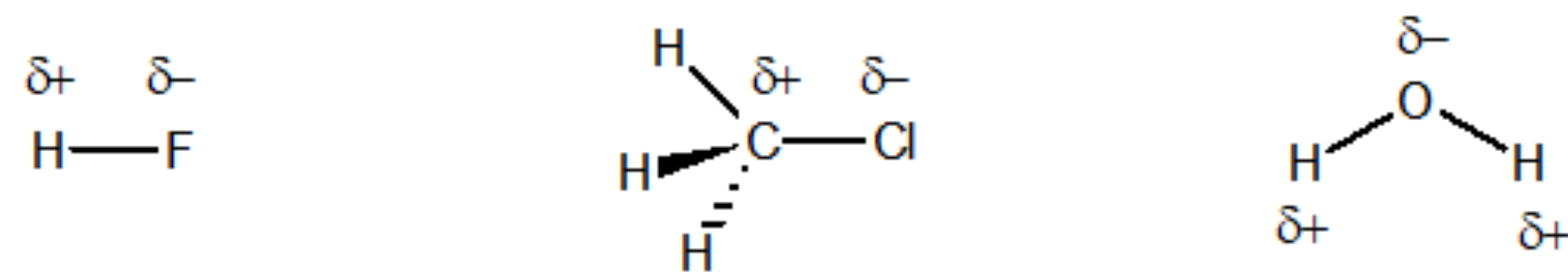


POLARITY

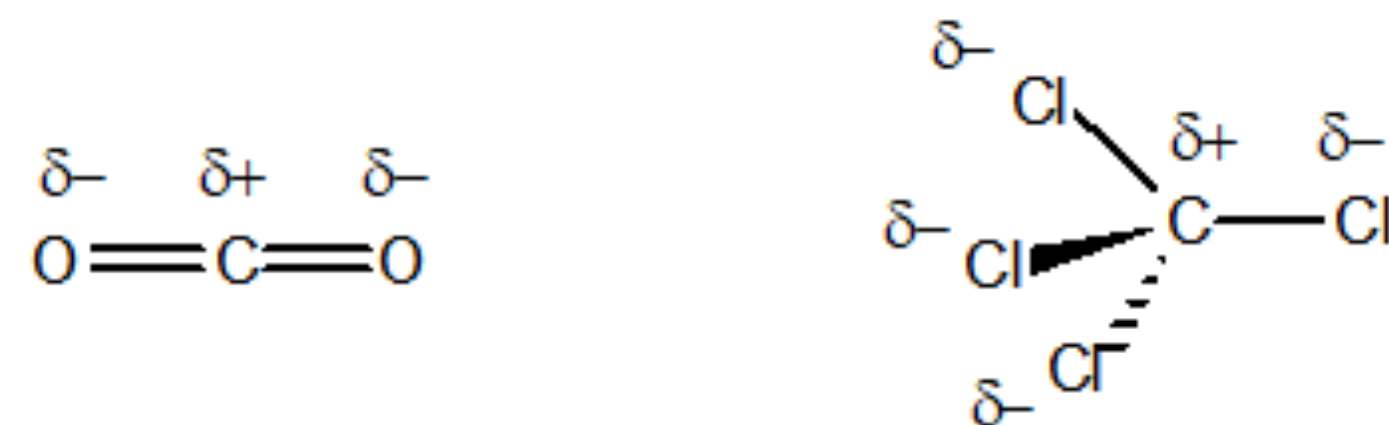


POLAR BONDS AND MOLECULES

- When the two atoms in a covalent bond have different electronegativities, the electrons are attracted more towards the more electronegative atom. This creates a slight charge separation with the more electronegative atoms being δ^- and the less electronegative atom being δ^+ . This charge separation creates a dipole and makes the bond polar.
- Many molecules with polar bonds are themselves polar and have a permanent dipole:



- However, some molecules with polar bonds do not have a permanent dipole as the bond polarities cancel each other out:

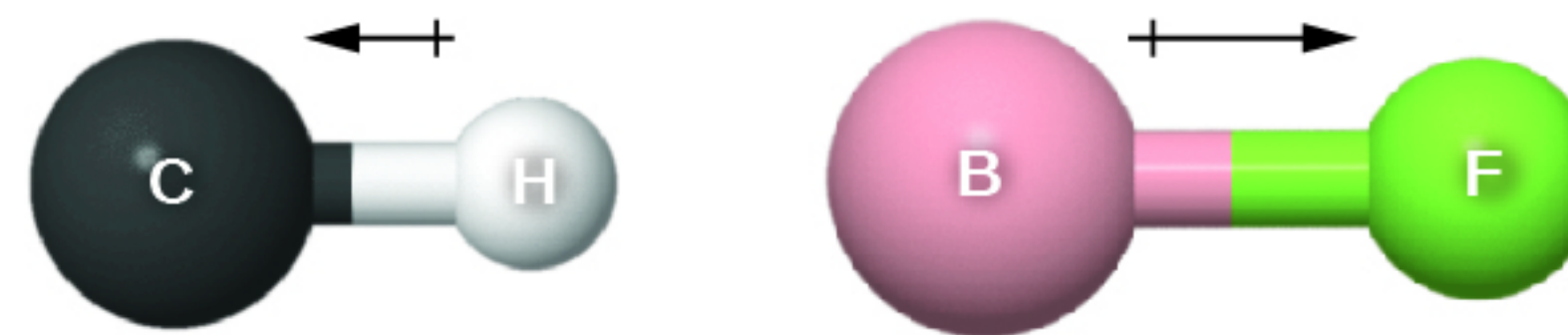


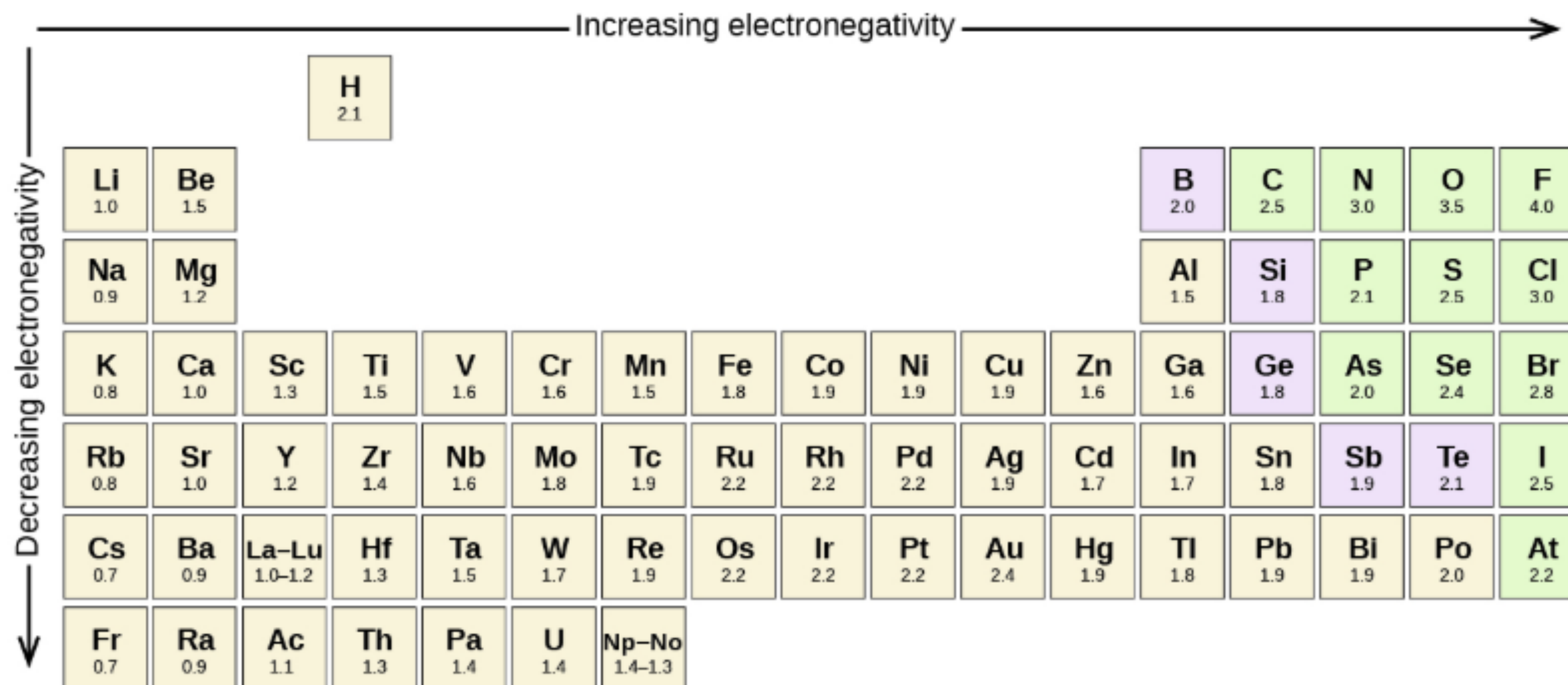
ELECTRONEGATIVITY AND POLAR BONDS

- Whether a bond is nonpolar or polar covalent is determined by a property of the bonding atoms called **electronegativity**.
- Electronegativity is a measure of the tendency of an atom to attract electrons (or electron density) towards itself. It determines how the shared electrons are distributed between the two atoms in a bond.
- The more strongly an atom attracts the electrons in its bonds, the larger its electronegativity. Electrons in a polar covalent bond are shifted toward the more electronegative atom; as such, the more electronegative atom is the one with the partial negative charge.
- The greater the difference in electronegativity, the more polarized the electron distribution and the larger the partial charges of the atoms.

ELECTRONEGATIVITY AND POLAR BONDS

- The figure on the next slide shows the electronegativity values of the elements as proposed by Linus Pauling.
- The charge separation which creates a dipole and makes the bond polar can be represented as a vector, a quantity having both direction and magnitude.
- Dipole vectors are shown as arrows pointing along the bond from the less electronegative atom toward the more electronegative atom. A small plus sign is drawn on the less electronegative end to indicate the partially positive end of the bond. The length of the arrow is proportional to the magnitude of the electronegativity difference between the two atoms.
- For example, there is a small difference in electronegativity between C and H, represented as a short vector below, whereas the electronegativity difference between B and F is much larger, so the vector representing the bond moment is much longer:



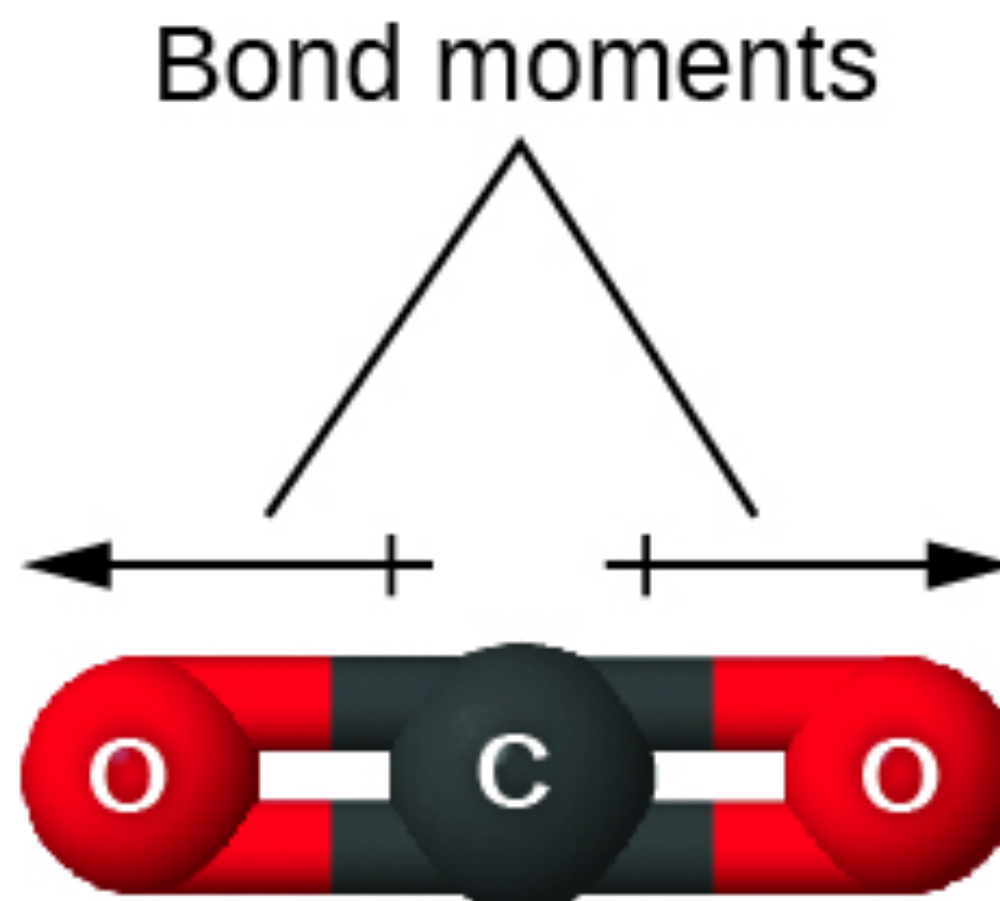


ELECTRONEGATIVITY AND POLAR BONDS

- A whole molecule may also have a separation of charge, depending on its molecular structure and the polarity of each of its bonds. If such a charge separation exists, the molecule is said to be a polar molecule (or dipole); otherwise the molecule is said to be nonpolar.
- The dipole moment measures the extent of net charge separation in the molecule as a whole. We determine the dipole moment by adding the bond moments in three-dimensional space, taking into account the molecular structure.
- For diatomic molecules, there is only one bond, so its bond dipole moment determines the molecular polarity.
- Homonuclear diatomic molecules such as Br_2 and N_2 have no difference in electronegativity, so their dipole moment is zero. For heteronuclear molecules such as CO , there is a small dipole moment. For HF , there is a larger dipole moment because there is a larger difference in electronegativity.

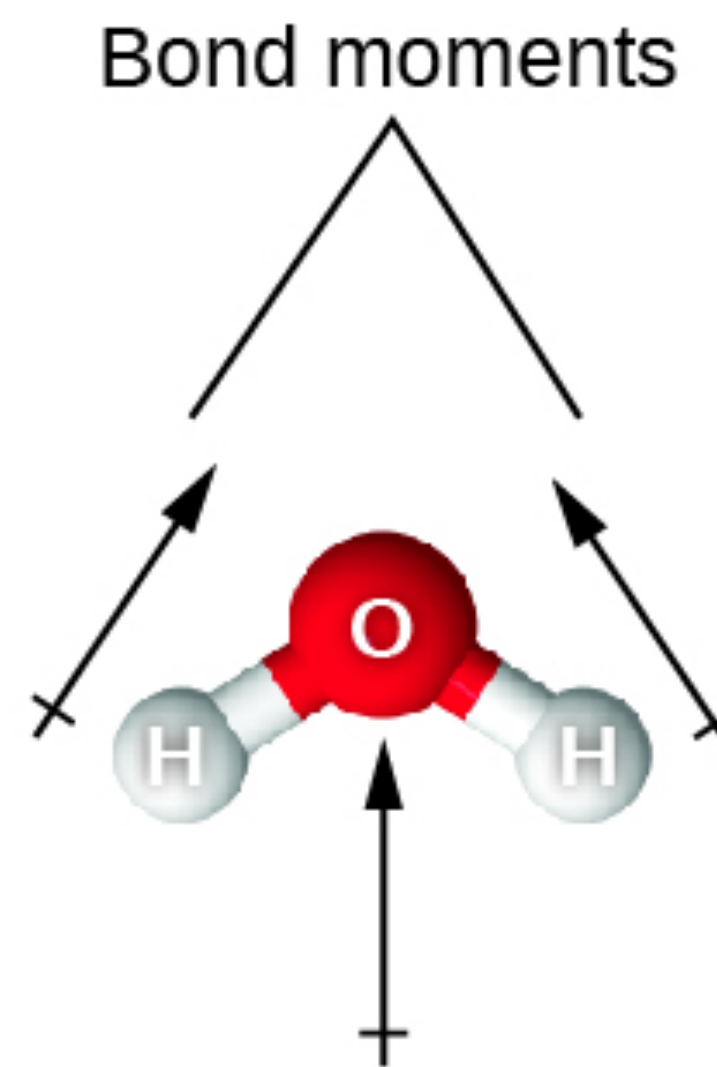
ELECTRONEGATIVITY AND POLAR BONDS

- When a molecule contains more than one bond, the geometry must be taken into account. If the bonds in a molecule are arranged such that their bond moments cancel (vector sum equals zero), then the molecule is nonpolar.
- This is the situation in CO_2 where each of the bonds is polar, but the molecule as a whole is nonpolar. From the Lewis structure, and using VSEPR theory, we determine that the CO_2 molecule is linear with polar $\text{C}=\text{O}$ bonds on opposite sides of the carbon atom. The bond moments cancel because they are pointed in opposite directions:



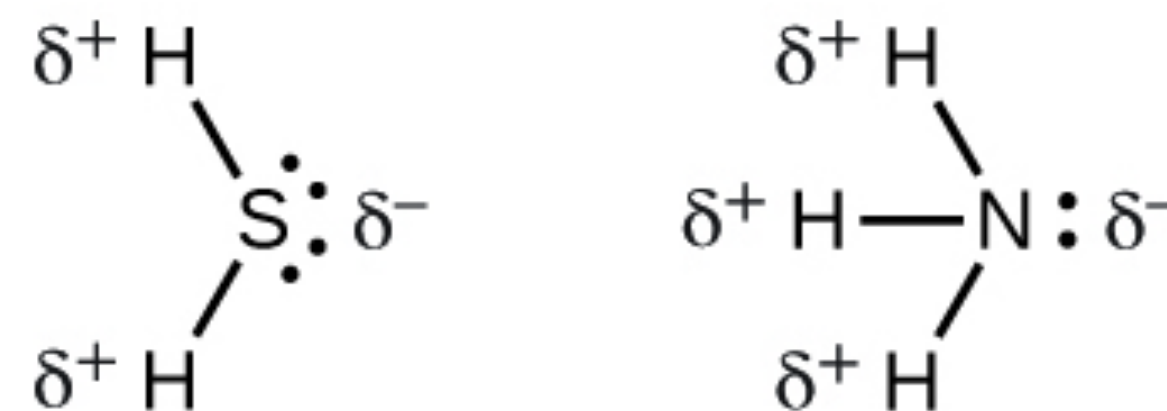
ELECTRONEGATIVITY AND POLAR BONDS

- In the case of the water molecule, the Lewis structure again shows that there are two bonds to a central atom, and the electronegativity difference again shows that each of these bonds has a nonzero bond moment.
- In this case, however, the molecular structure is bent because of the lone pairs on O, and the two bond moments do not cancel. Therefore, water does have a net dipole moment and is a polar molecule (dipole):



ELECTRONEGATIVITY AND POLAR BONDS

- When we examine the highly symmetrical molecules BF_3 (trigonal planar), CH_4 (tetrahedral), PF_5 (trigonal bipyramidal), and SF_6 (octahedral), in which all the polar bonds are identical, the molecules are nonpolar. The bonds in these molecules are arranged such that their dipoles cancel. However, just because a molecule contains identical bonds does not mean that the dipoles will always cancel. Many molecules that have identical bonds and lone pairs on the central atoms have bond dipoles that do not cancel.
- Examples include H_2S and NH_3 . A hydrogen atom is at the positive end and a nitrogen or sulfur atom is at the negative end of the polar bonds in these molecules:

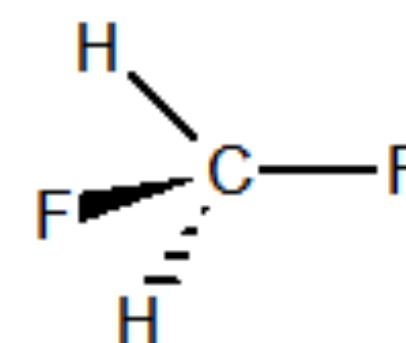
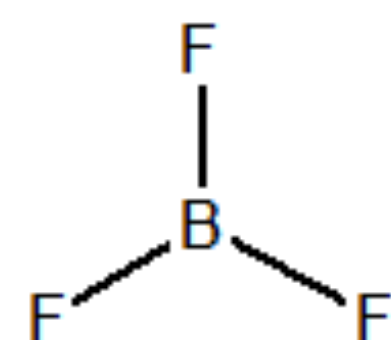
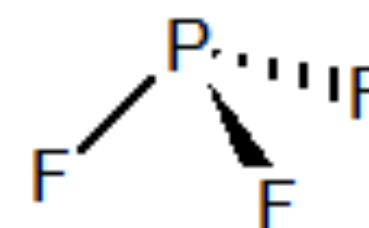
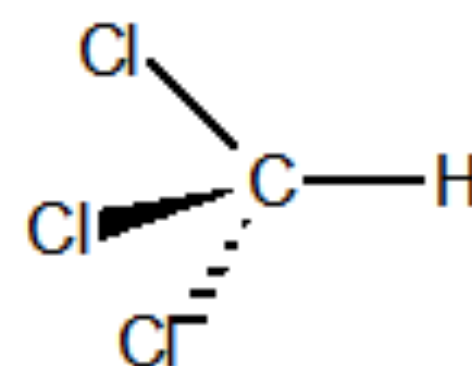


- To summarize, to be polar, a molecule must:
 - Contain at least one polar covalent bond.
 - Have a molecular structure such that the sum of the vectors of each bond dipole moment does not cancel.

POLAR BONDS AND MOLECULES

Example:

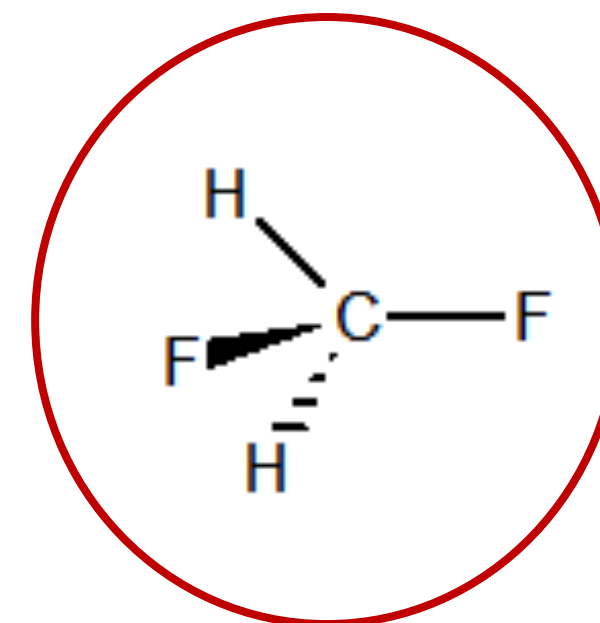
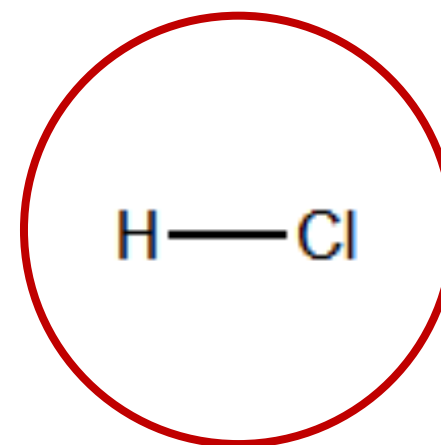
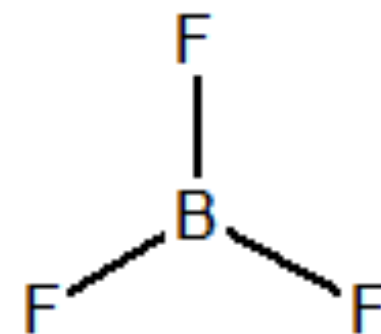
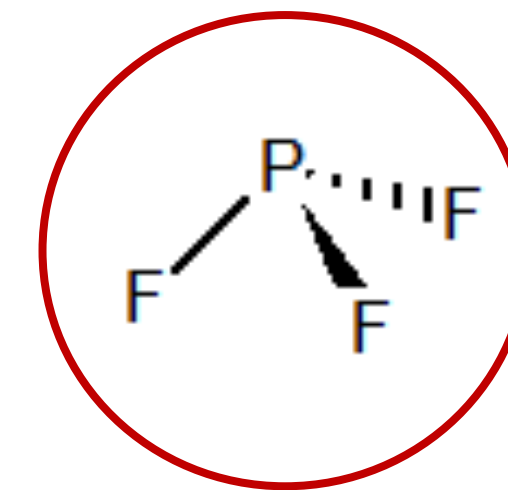
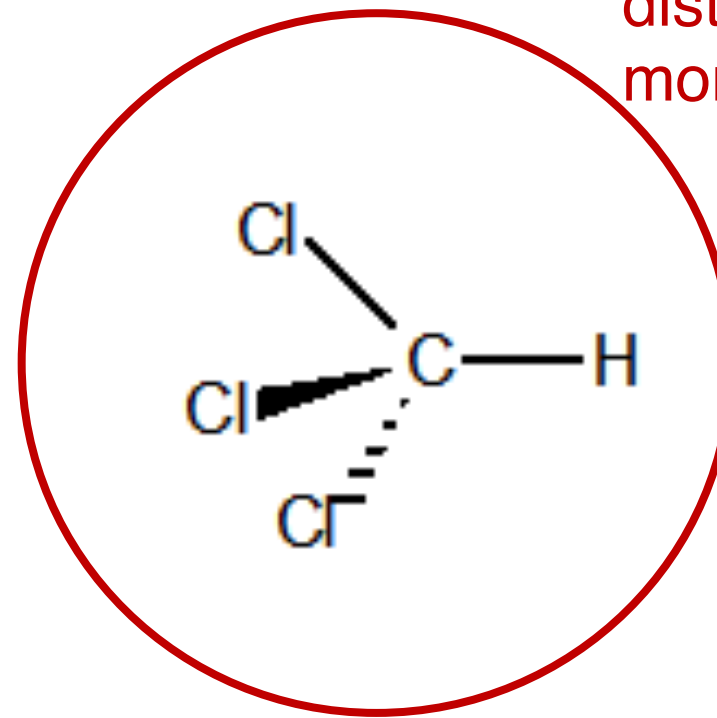
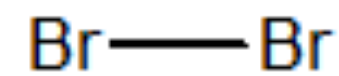
Which of the following molecules are polar?



POLAR BONDS AND MOLECULES

Example:

Which of the following molecules are polar?



CHCl3: There are two different bonds: C-H and C-Cl. Cl is more negatively charged than C and H. Therefore it does not have an even distribution of electronegativity, and one end is more negatively charged (Cl) than another (H)

PF3: the presence of a lone pair of electrons resulting in a bent structure due to electron-electron repulsion leads to an unequal distribution of charge within the molecule

HCl: Chlorine has a higher electronegativity than the hydrogen, so attracts electrons to spend more time at its end, giving it a negative charge and hydrogen a positive charge.

CH2F2: The pull of electrons towards the fluorine dipoles (i.e. the partial negative charge) and the resulting partial positive charges on the hydrogen ends of the tetrahedron results in dipole moments.

REREFENCES

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 2. Chemistry Library, The Living Library (https://chem.libretexts.org/Bookshelves/Introductory_Chemistry)
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